Polar Coordinates and Conversions

ANSWER KEY

Converting points

1 Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ $(2, 2\sqrt{3})$

b $\left(2, \frac{3\pi}{4}\right)$ $(-\sqrt{2}, \sqrt{2})$

2 Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$

b $(0, -5)$ $\left(5, \frac{3\pi}{2}\right)$

Converting equations

3 Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

Multiply by r: $r^2 = 6r\cos\theta$, so $x^2 + y^2 = 6x$, i.e. $(x-3)^2 + y^2 = 9$: a circle with center $(3, 0)$ and radius 3.

4 Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ $r = 4$

b $y = x$ $\theta = \frac{\pi}{4}$

5 Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

$r - r\sin\theta = 2$, so $r = y + 2$. Squaring: $x^2 + y^2 = (y+2)^2 = y^2 + 4y + 4$, so $x^2 = 4y + 4$ — a parabola.